Sahitya Kumar Choubey

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Software Engineering | *Full-Stack Development* | *Backend & AI Systems*

Technical Skills

- **Languages**: TypeScript, Python, JavaScript, SQL, C++
- **Frameworks and Libraries**: Node.js, Next.js, Express.js, React, Django, Django REST Framework, FastAPI, Tailwind CSS, WebSockets
- **Databases and ORMs**: PostgreSQL, MongoDB, Redis, SQLite, Weaviate (Vector DB), Prisma, Drizzle
- **Infrastructure and Tooling**: Docker, Docker Compose, Kubernetes, Apache Kafka, Puppeteer, Linux, Git, GitHub
- **Cloud Platforms**: Microsoft Azure, Google Cloud Platform (GCP), Vercel
- **AI/LLM Tooling**: LangGraph, LangChain, Model Context Protocol (MCP), Hugging Face

Experience

Software Engineering Intern Jan 2026 – Mar 2026

Memorly.AI Bangalore, Karnataka, India

- Designed a configurable, no-code test-automation platform for conversational agents to define, replay, and validate multi-step workflows, supporting 30+ agent deployments and 500+ automated test executions in place of repetitive manual scenario testing.
- Built 12+ schema-validated REST endpoints using Django REST Framework to synchronize inventory, pricing, address, and invoice data between a client's ERP system and an external marketplace across 15,000+ mapped products.
- Built pass/fail, latency, and feedback dashboards with structured feedback aggregation, reducing manual regression-testing effort by an estimated 85% across multiple client deployments.
- Tested, debugged, and shipped fixes that improved reliability of production conversational agents (WhatsApp ordering bots, feedback bots, call-based scheduling agents) across multiple clients, integrating Meta's WhatsApp Business API.

Research Intern May 2025 – Jul 2025

TCS Research & Innovation Hyderabad, Telangana, India

- Developed a FastAPI backend and Next.js frontend to deploy a LangGraph conversational agent for a user study, with real-time interaction and feedback capture (˜27 min average session).
- Designed a stateful LangGraph workflow (relevance validation, confidence-scored inference, adaptive follow-ups) across 6 financial-behavior dimensions, reducing manual annotation effort by an estimated 90%.
- Statistically analyzed 4,036 decisions from 92 Reddit users (chi-square, association-rule mining), showing value priorities differ by decision type ($p < .001$); co-authored two papers, one published at HCI International 2026 (DOI), one accepted at INDAM 2026.

Projects

Relay Source Code

Low-Latency, Horizontally Scalable Real-Time Chat System

- Built real-time messaging with WebSockets and Redis Pub/Sub fanout, sustaining 15,000+ concurrent connections across 3 replicas under load testing with 20–30ms p95 delivery latency.
- Architected an at-least-once Kafka ingestion pipeline with manual offset commits, achieving zero message loss across 200,000+ persisted messages while scaling producer throughput 31x via micro-batching.
- Improved sign-in throughput 4–5x by offloading password hashing off the Node.js event loop; shipped secure auth (Google OAuth, magic-link verification, HTTP-only cookie sessions).
- Validated horizontal scaling (2.4x HTTP throughput on 3 replicas) via Docker/Kubernetes stateless deployment, using a custom load-testing suite with CPU and TCP-queue profiling to isolate bottlenecks.

AutoCode Source Code

Terminal-Based Agentic Coding Assistant

- Built a multi-provider coding agent CLI from scratch: custom agent loop, SSE streaming, permission system, git-based undo, and MCP support, with 124 tests.
- Engineered retry handling for failed and cut-off model streams, holding 99.5% task completion with half of all requests failing in simulation (vs. 0.5% without retries) and no duplicated tool runs.
- Added context compaction cutting long sessions ˜90% with 65% key-fact recall (vs. 16.7% dropping oldest messages).
- Developed cross-session memory with hybrid retrieval (sqlite-vec, BM25, Gemini): 100% recall@5 vs. 66.7% keyword-only.
- Built eval harnesses (hidden checks, baselines, held-out tuning, confidence intervals) and reported where features did not help.

Leadership

Vice President & Webmaster Mar 2024 – May 2026

IEEE Computer Society & IEEE Student Branch, NIT Durgapur West Bengal, India

- Led web infrastructure for 12+ IEEE events (up to 200+ participants), maintaining event portals with a 60+ member team.

Education

National Institute of Technology, Durgapur West Bengal, India

Dual Degree (B.Tech + M.Tech) in Biotechnology, **CGPA: 8.24** Nov 2022 – May 2027

- **Relevant Coursework**: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Compiler Design, Machine Learning